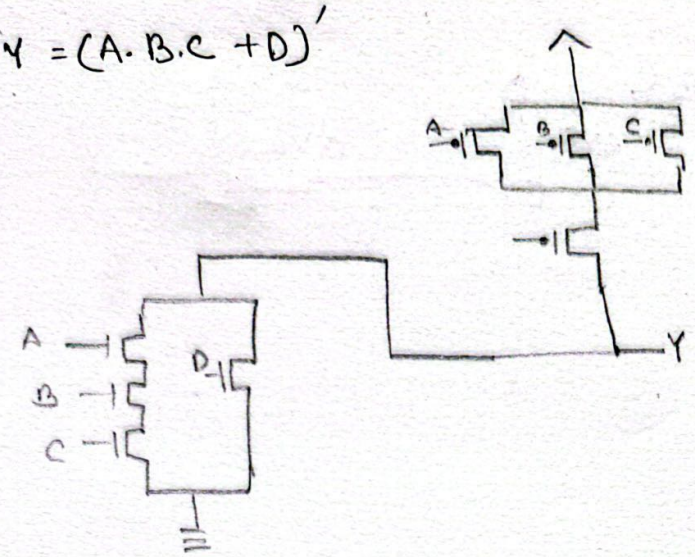
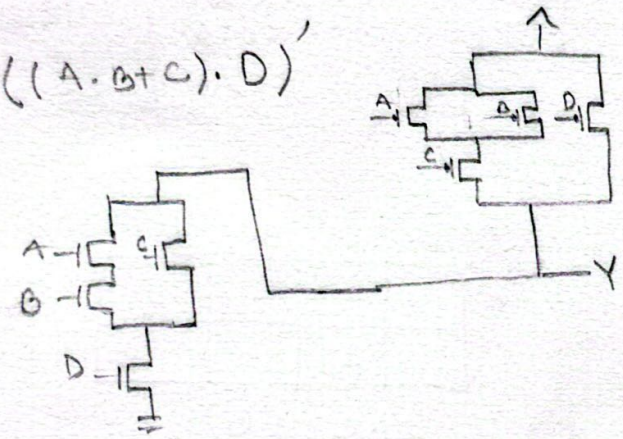
CSE460 Assignment-1

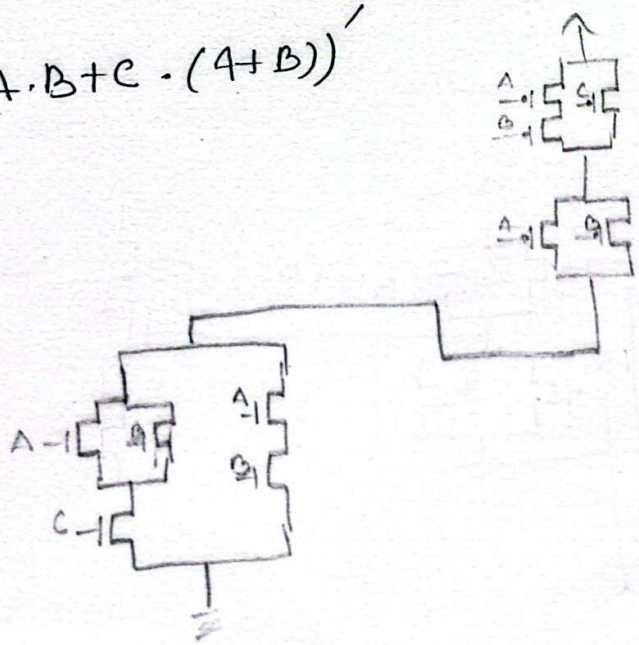
1 a) $Y = (A \cdot B \cdot C + D)'$



b) $Y = ((A \cdot B + C) \cdot D)'$



c) $Y \Rightarrow (A \cdot B + C \cdot (A + B))'$



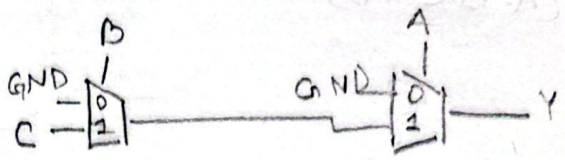
Scanned with CamScanner

2

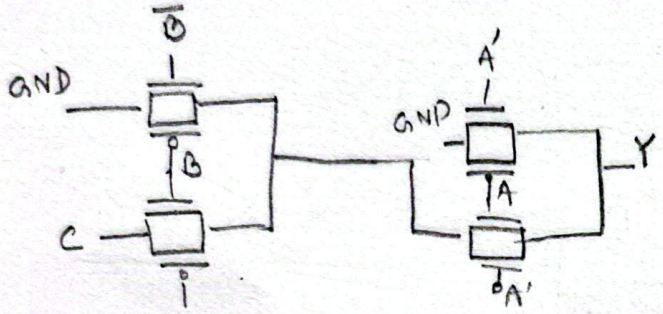
a

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$F = A \cdot B \cdot C$



b



3

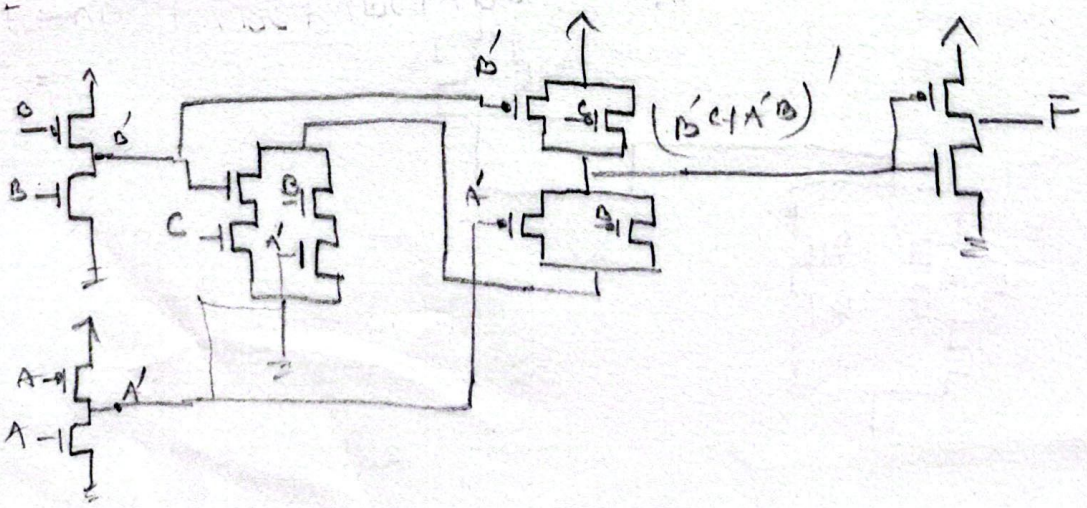
a

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

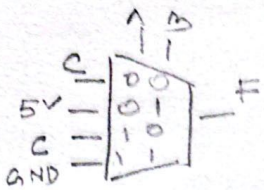
A \ BC	B'C'	B'C	BC	BC'
A'	0	1	1	1
A	0	1	1	0

$F = (B'C + A'B)'$

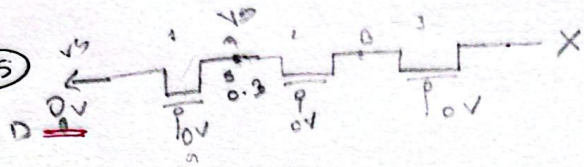
b



(2)



(5)



for 1

$V_D = 0V$

$V_{SG} \geq |0.3|$

$V_S - 0 \geq 0.3$

$V_S = 0.3$

for 2

$V_D = 0.3V$

$V_{SG} \geq |0.3|$

$V_S - 0 \geq 0.3$

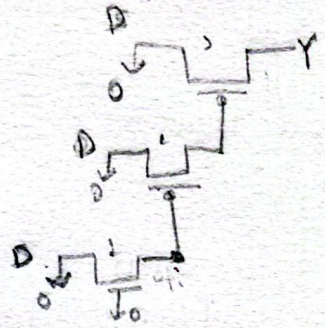
$V_S = 0.3$

for 3

$V_D = 0.3V$

$V_{SG} \geq |0.3|$

$V_S = 0.3V = X$



for1

$V_{SG} \geq |0.3|$

$V_S - 0 \geq 0.3$

$V_S \geq 0.3$

$V_S = 0.3$

for2

$V_G = 0.3$

$V_{SG} \geq |0.3|$

$V_S - 0.3 \geq 0.3$

$V_S = 0.6$

for 3

$V_G = 0.6$

$V_{SG} \geq |0.3|$

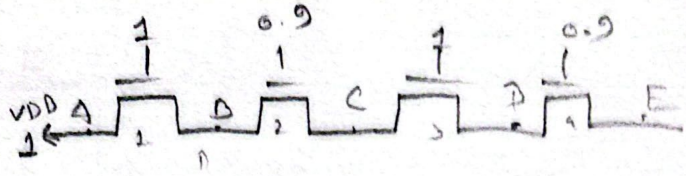
$V_S - 0.6 \geq 0.3$

$V_S = 0.9 = Y$

X & Y differs cause X has parallel PMOS where Y has series

Scanned with CamScanner

(8)



$V_{dd} = 1V$
$V_{gnd} = 0$
$V_{tn} = 0.2$

For 1 $V_D = 1$

$V_{GS} \geq V_t$
$1 - V_S \geq 0.2$
$V_S \leq 0.8$
$B = 0.8$

for 2 $V_D = 0.8$

$V_{GS} \geq 0.2$
$0.9 - V_S \geq 0.2$
$V_S \leq 0.7$
$C = 0.7$

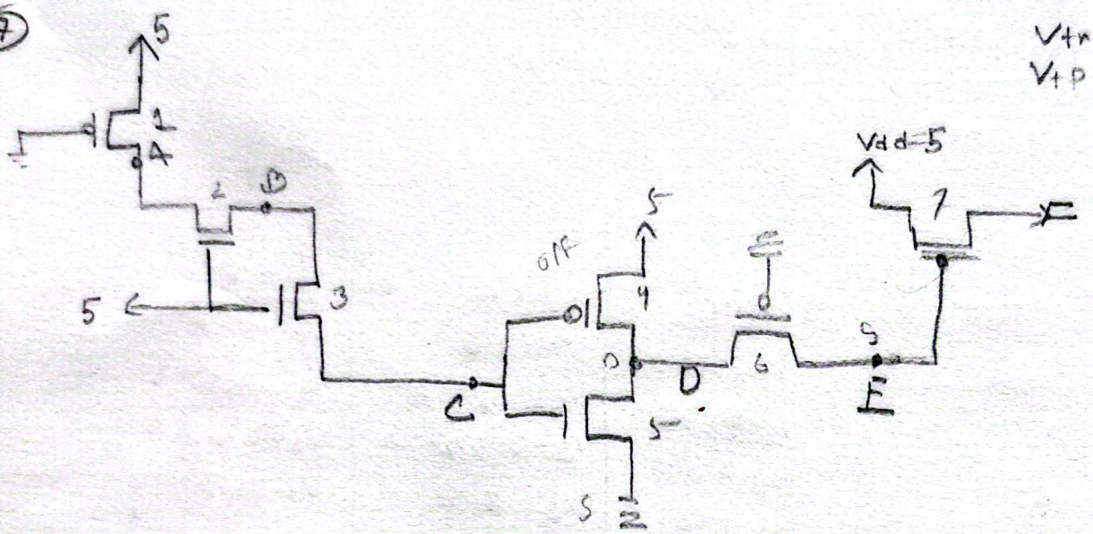
for 3 $V_D = 0.7$

$1 - V_S \geq 0.2$
$V_S \leq 0.8$
$D = 0.7$

for 4 $V_D = 0.7$

$0.9 - V_S \geq 0.2$
$V_S \leq 0.7$
$E = 0.7$

(7)



$V_{tn} = 0.5$
$V_{tp} = 1.5$

for 1:

$A = 5V$

for 2

$V_{GS} > 0.5$
$5 - V_S \geq 0.5$
$V_S \leq 4.5$
$B = 4.5V$

for 3

$5 - V_S \geq 0.5$
$V_S \leq 4.5$
$C = 4V$

for 5

AS, $V_S = 0V$
$V_D = 0V$
$D = 0V$

for 6

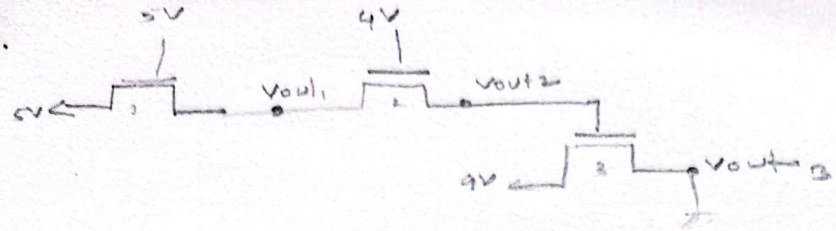
$V_{SG} \geq 1.5$
$V_S \leq 1.5$
$E = 1.5V$

for 7

$V_{SG} > 1.5$
$5 - 1.5 \geq 1.5$
$3.5 > 1.5$
$F = 5V$

Scanned with CamScanner

8.



$V_{in} = 0.5$

for 1

$V_{GS} \geq 0.5$

$5 - V_S \geq 0.5$

$V_S \geq 4.5$

$V_{out1} = 4.5$

for 2

$V_{GS} \geq 0.5$

$4 - V_S \geq 0.5$

$V_S \geq 3.5$

$V_{out2} = 3.5$

for 3

$V_{in} = 3.5$, $V_D = 4$

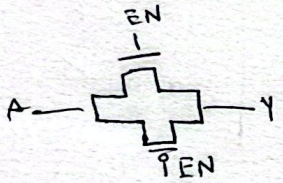
$V_{GS} \geq 0.5$

$3.5 - V_S \geq 0.5$

$V_S \geq 3$

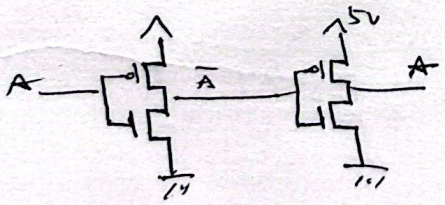
$V_{out3} = 3$

4. <u>Non restoring property</u>: Whe the input has noise ~~th~~ it can pass through the device to output, Such as transistors buffer



Noise in A can pass to Y

Restoring property of a device is reverse of nonrestoring property. Noise of input will not pass to the output because the output high comes from V_{DD} & low output comes from GND.



Noise cant pass

Scanned with CamScanner